

MICA: Can an AI Understand What a Minecraft Builder Is Doing?

Minecraft Intention-Aware Collaborative Agent · Benjamin [surname], [institution] · recognizing situations and intentions from raw observation — no instructions, no privileged game state

Why this matters

In structured work — construction, warehouses, assembly — an assistant that must be **told everything is bottlenecked by the telling**. The scarce resource is the human’s attention. An assistant that *infers* what you are doing removes that bottleneck.

Assistance-game research formalized this (CIRL, 2016) and proved goal inference is the load-bearing part: a **confidently wrong helper is worse than none** (Watch-And-Help: 40% of episodes, the helper undid the human’s work).

The gap: every prior assistant reads oracle game state. **Robots don’t get oracle state**. MICA asks the question under robot conditions — perception in, calibration measured, authority earned.

The pipeline

proof-grade capture
every block event (actor-attributed), POV, input state

frozen 2D + 3D perception
VPT behavior embedding · MineCLIP per-goal similarity · Uni3D shape

Bayesian belief $b(\text{goal, mode})$
learned likelihood heads · do-operator: agent’s own blocks are never evidence

safety gate + FSM
observe < suggest < preview < place · hysteresis · vetoes

embodied agent
follows, scans, voices offers · places ONE block only on your chat “yes”

Every model is **frozen**: models built to make agents *act* (VPT, MineCLIP) are read backwards here, as *evidence about a human*.

The dataset (all real, all mine)

labeled real free-build sessions	37
matcher-agreed / contested / discarded	10 / 18 / 9
never-seen evaluation sessions	24
held-out real rows (anticipation eval)	1,158
training pairs (one reproducible cascade)	12,045
scripted known-truth corpus	120 builds

One command regenerates every number on this poster from raw recordings.

METHOD — how every number here was made trustworthy

Pre-registered. Criteria, decision rules, and holdouts frozen and dated *before* every run — and they fired: one verdict reversed, one table retracted, failures banked.

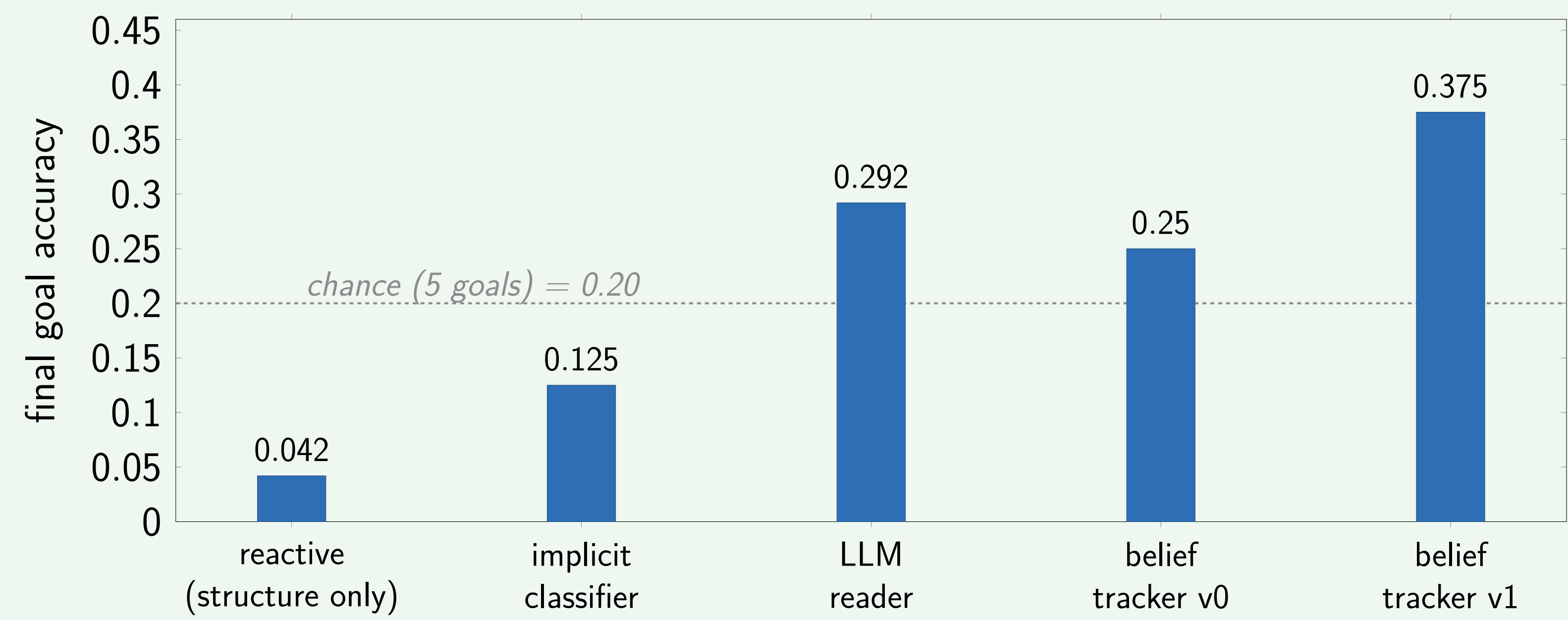
Known-truth anchored. A scripted corpus whose labels are true by construction: deterministic components are *measured* (matcher 1.0), never assumed.

Leakage rules, standing. Evidence strictly precedes the action it scores; the agent’s own blocks are never evidence (do-operator); violations are asserts, not reviews.

Controlled arms. Four arms differ *only* in the intent mechanism — same gate, same exposure-verified splits; experiments train to separate files, production checksummed.

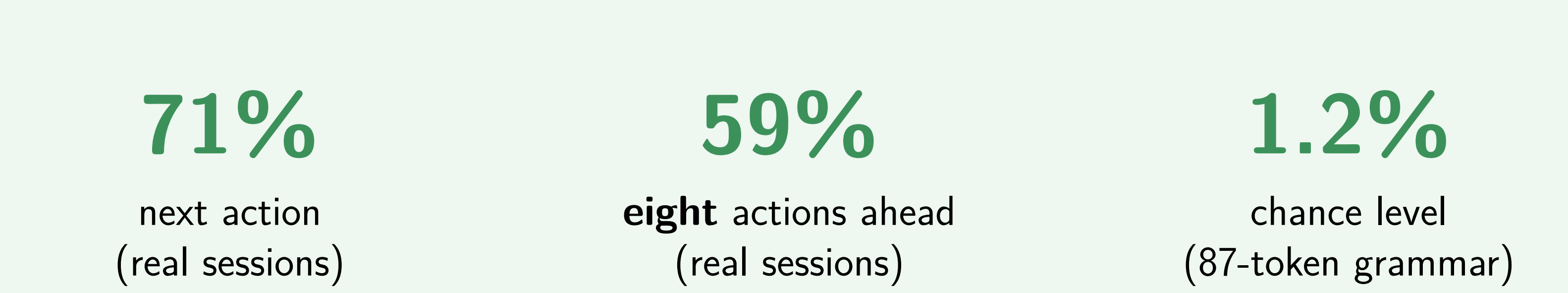
One-command reproducible. A single ledger-stamped cascade regenerates every number on this poster from the raw recordings.

Finding 1 — Intent inference is NECESSARY off-template



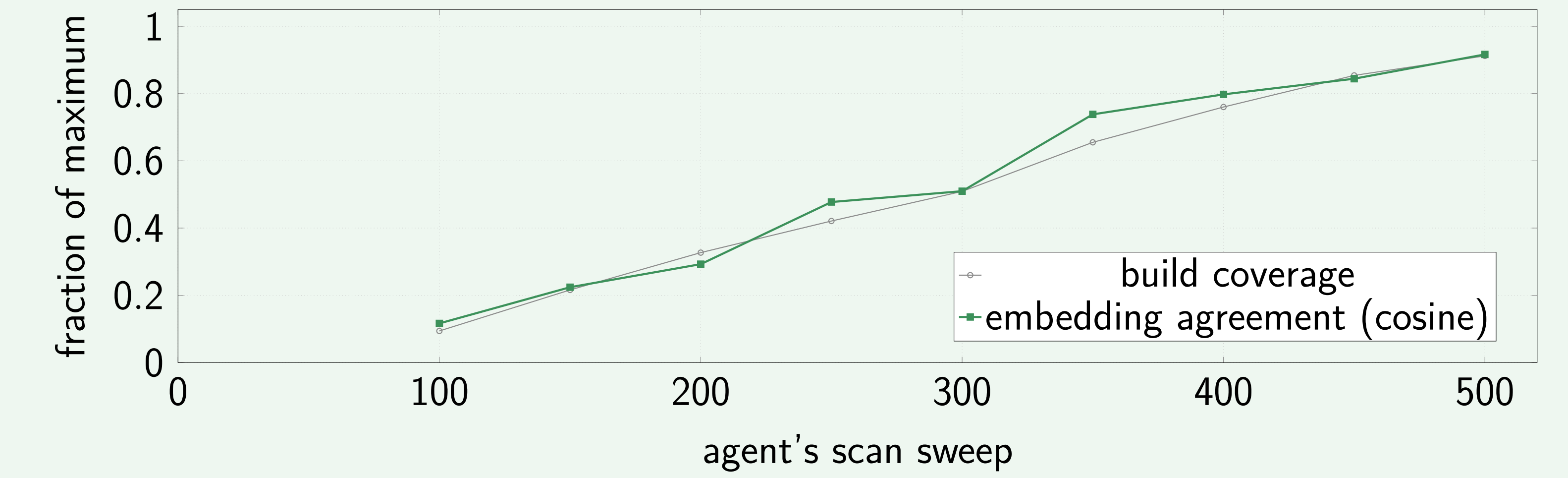
On **24 real sessions no model ever trained on**, the helper that skips intent inference collapses to **0.042** — a $9\times$ gap to the belief tracker. Exactly where builds stop matching templates, structure lookup stops working. *Intent inference is what survives*.

Finding 2 — It anticipates your next moves, with minimal context



No instructions, no goal ever specified, frozen perception only. And the training data was free: **label-free pretraining on real sessions** closed a 1.25-nat domain gap (−37% prediction error) — contested and discarded sessions included, because no label enters the sample.

Finding 3 — Understanding accumulates with observation



As the embodied agent sees more of the build, the shape embedding of *only what it has seen* converges to the full structure’s **(0.92 cosine at 91% coverage)**.

What failed — and why that is the point

This project pre-registered its criteria and let them fire:

- ▶ **Early recognition: unproven.** The pre-registered early lock-in test fails for every arm ($p = 0.06\text{--}1.0$). Recognition is mid-build, not early — more sessions are the path.
- ▶ **Confidence: not calibrated.** Above 0.4 stated confidence, held-out reliability *inverts*. Consequence taken seriously: the confidence-licensed placement route is disabled by its own threshold.
- ▶ **The belief is not prediction fuel.** Forcing the belief into the action decoder changes nothing ($|\Delta| \leq 0.001$ at every horizon) — its value is *judgment* (what to say, when to act), not forecasting.
- ▶ **The agent has never placed a block.** Zero, ever — pinned. Assistance here is *designed and safety-gated, never evaluated*. A helper that declines to act without license is the safety design working.
- ▶ **One verdict was reversed** — by finding a feature leak, repairing the corpus, and re-running the pre-registered experiment (it then won on every criterion). The retraction is on the record.

The system improves lawfully with the right data

Two pre-registered experiments rode one retrain; both won; both adoptions were separate, dated decisions:

	control	adopted
real-session prediction error (NLL)	3.42	2.17
clean-eval accuracy (+ contested labels)	0.00	0.33
calibration error (ECE)	0.258	0.232
earliness (sustained-from, lower=earlier)	0.990	0.897

What this means beyond Minecraft

The approach transfers to any domain with: **(1)** discrete observable actions, **(2)** goals legible from partial structure, **(3)** cheap dense observation of a shared workspace — bricklaying, timber framing, warehouse picking, lab bench work.

Next: recalibration (the gate to autonomy) · consent-route sessions with *placement survival* as the outcome measure · more sessions for the earliness question.

Take one sentence home

*“A helper that only pattern-matches the environment dies the moment work deviates from templates — and real work always deviates.
Inferring intention is what survives.”*